



REFLECTIVE REPORT
FOR
BOARDER ASSIST

**AI-Powered Personalized Tourism Recommendation
System for Sri Lanka**

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1.Introduction

1.1 Background

Tourism is one of the major contributors to the Sri Lankan economy. Thousands of local and international tourists visit the country every year to experience its cultural heritage, natural attractions, wildlife and beaches. However, most existing tourism platforms provide the same recommendations to every user without considering individual preferences. This makes it difficult for travellers to quickly identify destinations, hotels and activities that best match their interests.

This project focuses on developing an intelligent tourism recommendation system that provides personalised recommendations based on user preferences and tourism information. The system combines modern web technologies with machine learning techniques to improve recommendation accuracy and enhance the overall travel planning experience. In addition, tourism demand forecasting is incorporated to support better recommendations during different travel seasons.

1.2 Motivation

I selected this project because I wanted to improve my knowledge of software engineering while learning machine learning and recommendation systems. Tourism is an important industry in Sri Lanka, and I believed that developing a personalized recommendation platform would provide practical value to both travelers and tourism businesses.

This research also allowed me to strengthen my skills in data preprocessing, model development, backend API development and modern frontend technologies. I believe the experience gained through this project will be valuable for my future career as a Computer Science engineer.

1.3 Overview of the report

This report reflects on my experiences throughout the development of the project. It discusses requirement gathering, literature review, dataset preparation, machine learning model development, backend and frontend implementation, testing, challenges encountered, lessons learned and future improvements. The report also explains how these activities contributed to my technical and professional development.

2. Self-Management

Managing this research project required careful planning and continuous learning. I divided the work into several stages, including requirement gathering, literature review, dataset preparation, machine learning development and system implementation. This helped me manage my time effectively while monitoring the progress of the project.

Throughout the research, I regularly discussed my progress with my supervisor and incorporated the feedback received into the project. During the machine learning phase, I encountered challenges related to data quality and model performance. Instead of rushing through the work, I spent additional time improving the dataset and comparing multiple algorithms before selecting the final model.

The project also encouraged me to learn several new technologies independently, including FastAPI, TensorFlow, XGBoost and Next.js. These experiences improved my problem-solving abilities, technical knowledge and confidence in handling a research-based software engineering project.

3. Reflection

3.1 Requirement Gathering

The first stage of the project involved understanding the problem and identifying the system requirements. I analyzed several existing tourism websites and recommendation platforms to understand their strengths and limitations. I also discussed the project scope with my supervisor to identify the essential functional and non-functional requirements.

One of the main challenges during this phase was defining the project scope. Initially, I planned to include many advanced features, but after several discussions, I refined the scope to focus on personalized recommendations, tourism demand forecasting and a user-friendly web platform. This helped me keep the project realistic while maintaining its research value.

Through this process, I understood the importance of proper requirement gathering. A clear understanding of the project objectives helped minimize unnecessary changes during later development stages.

3.2 Literature Review

The literature review helped me understand how recommendation systems have evolved and how machine learning techniques are applied within the tourism industry. I studied research

papers related to content-based filtering, hybrid recommendation systems and tourism demand prediction.

One challenge was comparing different research approaches, as each study used different datasets and evaluation methods. To overcome this, I focused on identifying common techniques and understanding their advantages and disadvantages rather than directly comparing their results.

This activity improved my research skills and provided a strong foundation for selecting suitable technologies and algorithms for my own project.

3.3 Dataset Collection

Collecting the dataset was one of the most time-consuming stages of the project. The data was gathered from multiple tourism-related sources, including hotels, restaurants, tourist attractions, activities and tourism statistics. After combining these sources, the final dataset contained more than 51,000 records.

The biggest challenge was maintaining data consistency because different sources used different formats and naming conventions. I standardized the data structure before moving to the preprocessing stage.

This experience showed me that the quality of the dataset directly affects the quality of machine learning models.

3.4 Data Cleaning

The collected dataset contained duplicate records, missing values and inconsistent information. Therefore, I performed several preprocessing tasks, including removing duplicate entries, handling missing values and standardizing categorical values.

Although this process required considerable time, it significantly improved the overall quality of the dataset. I realized that data cleaning is one of the most important stages in any machine learning project because even a powerful algorithm cannot produce reliable results from poor-quality data.

3.5 Feature Engineering

Feature engineering was carried out to improve the effectiveness of the recommendation model. Textual information was converted using TF-IDF, while numerical features were scaled and categorical values were encoded before training the models.

Initially, I experimented with several feature combinations, but some of them reduced model performance. After multiple experiments, I selected the features that produced the best results while avoiding unnecessary complexity.

This stage taught me that selecting the right features is often more important than simply choosing a complex algorithm.

3.6 Machine Learning Development

During this project, I implemented and evaluated several machine learning models, including KNN, SVM, Random Forest, AdaBoost, XGBoost, LSTM and hybrid models. Each model was trained and evaluated using Accuracy, Precision, Recall and F1-score.

Some early experiments produced unrealistic results due to feature leakage, which required me to redesign the preprocessing pipeline. After correcting the dataset and tuning the model parameters, the evaluation became more reliable.

Among all the models, XGBoost achieved the highest performance with an accuracy of 99.97%, making it the most suitable algorithm for generating personalized recommendations. LSTM was selected separately for tourism demand forecasting because of its ability to learn temporal patterns from sequential data.

Comparing multiple models helped me understand that selecting the most appropriate algorithm requires careful experimentation rather than relying solely on published research.

3.7 Backend Development

The backend was developed using Express.js together with FastAPI. Express.js manages the application services, while FastAPI handles the recommendation engine and machine learning predictions. MongoDB is used to store tourism information and user data.

During development, I encountered issues related to API communication and data formatting between the frontend and the machine learning service. These issues were resolved by improving API validation and response handling.

Working on the backend improved my understanding of API development and system integration.

3.8 Frontend Development

The frontend was developed using Next.js, React and Tailwind CSS to provide a modern and responsive user interface. Separate dashboards were designed for tourists, hotel owners, activity providers and administrators.

One challenge was designing interfaces that remained simple while displaying a large amount of tourism information. After several design improvements, I created a cleaner layout that enhances usability across different devices.

This phase strengthened my frontend development skills and improved my understanding of user experience design.

3.9 Testing

Testing was performed throughout the development process to ensure the reliability of both the software system and the machine learning models. The recommendation models were evaluated using standard performance metrics, while APIs were tested to verify correct data exchange between system components.

Frontend testing was also conducted to ensure responsive layouts and correct navigation across different user roles. Integration testing confirmed that the frontend, backend and recommendation engine worked together successfully.

Overall, testing helped identify several issues during development and improved the stability and reliability of the final system.

4. Current Progress

At the current stage, the major components of the project have been successfully completed. The requirement gathering, literature review, dataset collection and preprocessing phases have been finalized. The machine learning models have been trained and evaluated, and XGBoost was selected as the final recommendation model due to its superior performance. LSTM has been incorporated for tourism demand forecasting.

The backend has been developed using Express.js and FastAPI, while the frontend has been implemented using Next.js and Tailwind CSS. Separate dashboards have been created for different user roles, including tourists, hotel owners, activity providers and administrators. The remaining work mainly focuses on improving the recommendation interface, integrating booking functionality and conducting user evaluation before the final submission.

5. Good Practices

Throughout this project, I followed several good software engineering practices that helped improve the quality of the final system. I maintained my source code using GitHub, allowing

me to track changes and manage different development stages efficiently. Writing modular and reusable code made future modifications easier and reduced unnecessary duplication.

I also documented my work regularly, which helped me monitor progress and simplify future improvements. During machine learning development, I carefully evaluated multiple models before selecting the final solution instead of relying on a single algorithm. Continuous testing during development also helped identify issues early and improved the overall stability of the system.

6. Mistakes and Changes

During the project, I encountered several challenges that required changes to my original approach. Initially, the collected dataset contained duplicate records, inconsistent values and missing information, which affected the quality of the machine learning models. After identifying these issues, I spent additional time cleaning and preparing the dataset before retraining the models.

Another challenge occurred during model evaluation when some experiments produced unrealistically high results due to feature leakage. After investigating the preprocessing pipeline, I corrected the issue and repeated the training process using properly separated datasets.

I also made several improvements to the frontend design after receiving feedback from my supervisor. Simplifying the interface improved usability and made the system easier for users to navigate.

These experiences taught me that mistakes are a natural part of the development process and that carefully analyzing problems leads to better solutions.

7. Lessons Learned

This project has significantly improved my technical and personal skills. I gained practical experience in machine learning, recommendation systems, backend API development and modern frontend technologies. Working with a large dataset also strengthened my understanding of data preprocessing and feature engineering.

Beyond technical knowledge, I learned the importance of planning, communication and continuous learning. Regular discussions with my supervisor helped improve both the research methodology and system design. I also developed better problem-solving skills by analyzing issues independently and identifying suitable solutions.

Overall, this research has increased my confidence in developing large-scale software engineering projects while improving my research and analytical abilities.

8. If I Got Another Chance

If I had the opportunity to complete this project again, I would begin collecting and preparing the dataset much earlier because this stage required more time than I initially expected. I would also conduct user surveys during the early stages of development to gather additional requirements and validate the recommendation process.

In addition, I would explore deep learning recommendation techniques and include more real-time tourism information to improve recommendation accuracy. These improvements would further enhance both the research contribution and the practical value of the system.

9. Future Work

Although the current system successfully provides personalized tourism recommendations, several enhancements can be implemented in future versions. Booking functionality can be integrated to allow users to reserve hotels and activities directly through the platform. Explainable recommendation features can also be introduced to help users understand why particular destinations are recommended.

Future work may also include developing a mobile application, collecting user feedback to continuously improve recommendations and integrating real-time tourism information such as weather, traffic conditions and seasonal events. These enhancements would make the system more intelligent and practical for real-world deployment

10. Detailed Thesis Contents

Title Page

(Includes the University of Bedfordshire logo, project title, student name, student ID, supervisor, and submission details.)

Abstract

(Provides a summary of the research problem, objectives, methodology, implementation, and key findings.)

Acknowledgements

(Expresses gratitude to the supervisor, lecturers, family, and everyone who supported the project.)

Table of Contents

List of Figures

List of Tables

List of Abbreviations

Chapter 1: Introduction

1.1 Project Background

(Introduces the research problem, motivation, and the need for an AI-powered tourism recommendation system.)

1.2 Research Aim and Objectives

(Presents the overall aim and objectives of the research.)

1.3 Project Realization

(Provides a brief overview of the project development process.)

1.4 Structure of the Report

(Summarizes the contents of each dissertation chapter.)

Chapter 2: Literature Review

2.1 Tourism Recommendation Systems

Reviews existing tourism recommendation approaches.

2.2 Artificial Intelligence in Tourism

Discusses AI and Machine Learning applications in tourism.

2.3 Recommendation Models

Reviews TF-IDF, KNN, SVM, Random Forest, AdaBoost, XGBoost, and LSTM.

2.4 Similar Systems

Compares existing tourism recommendation systems.

2.5 Research Gap

Identifies limitations in existing systems and the contribution of the proposed solution.

Chapter 3: Methodology

3.1 Research Methodology

Explains the research methodology and development process.

3.2 Project Planning

Describes project planning and the Gantt chart.

3.3 Requirement Gathering and Analysis

Explains requirement collection and system analysis.

3.4 System Design

Presents the system architecture, database, and design diagrams.

3.5 Implementation and Testing

Describes implementation, model development, and system testing.

Chapter 4: Results and Discussion

Presents the machine learning results, system outputs, testing results, screenshots, and discusses the overall findings.

Chapter 5: Evaluation

Evaluates the developed system, compares model performance, and discusses the strengths and limitations.

Chapter 6: Conclusion

6.1 Conclusion

Summarizes the research and overall project outcomes.

6.2 Limitations

Discusses the limitations encountered during the project.

6.3 Future Work

Describes possible future improvements, including Explainable AI, mobile application support, real-time data integration, and enhanced recommendation models.

References

Lists all references using the Harvard Referencing Style.

Appendices

Appendix A – System Screenshots

Appendix B – Project Gantt Chart

Appendix C – Machine Learning Model Comparison

Appendix D – System Architecture Diagram

Appendix E – Database Design

Appendix F – REST API Documentation

Project Gantt Chart

